

# Phosphate Additives in Food—a Health Risk

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**Background** Hyperphosphatemia has been identified in the past decade as a strong predictor of mortality in advanced chronic kidney disease (CKD). For example, a study of patients in stage CKD 5 (with an annual mortality of about 20%) revealed that 12% of all deaths in this group were attributable to an elevated serum phosphate concentration. Recently, a high-normal serum phosphate concentration has also been found to be an independent predictor of cardiovascular events and mortality in the general population. Therefore, phosphate additives in food are a matter of concern, and their potential impact on health may well have been underappreciated.

**Methods** We reviewed pertinent literature retrieved by a selective search of the PubMed and EU databases ([www.zusatzstoffe-online.de](http://www.zusatzstoffe-online.de), [www.codexalimentarius.de](http://www.codexalimentarius.de)), with the search terms “phosphate additives” and “hyperphosphatemia.”

**Results** There is no need to lower the content of natural phosphate, i.e. organic esters, in food, because this type of phosphate is incompletely absorbed; restricting its intake might even lead to protein malnutrition. On the other hand, inorganic phosphate in food additives is effectively absorbed and can measurably elevate the serum phosphate concentration in patients with advanced CKD. Foods with added phosphate tend to be eaten by persons at the lower end of the socioeconomic scale, who consume more processed and “fast” food. The main pathophysiological effect of phosphate is vascular damage, e.g. endothelial dysfunction and vascular calcification. Aside from the quality of phosphate in the diet (which also requires attention), the quantity of phosphate consumed by patients with advanced renal failure should not exceed 1000 mg per day, according to the guidelines.

**Conclusion** Prospective controlled trials are currently unavailable. In view of the high prevalence of CKD and the potential harm caused by phosphate additives to food, the public should be informed that added phosphate is damaging to health. Furthermore, calls for labeling the content of added phosphate in food are appropriate.